

Photorealistic 3D Holographic Display with Natural Defocus Effect

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Holographic displays deliver realistic visual experiences by reconstructing three-dimensional (3D) light fields. However, widely used RGB-D data offers reliable geometry only in focused regions, lacking light field information in defocused areas, particularly with transparent or reflective surfaces. Existing methods typically utilize coherent diffraction to approximate defocus effects in incoherent perception, introducing ringing artifacts and unnatural depth transitions. We propose a method generating physically consistent complex-valued light fields from multi-layer RGB-D data. Our approach integrates neural rendering and multi-branch phase prediction networks to compute amplitude and phase, while explicitly enforcing wave propagation to preserve coherence and physical consistency. Our simulations achieve an average peak signal-to-noise ratio of 32.35 dB and an average structural similarity index of 0.938. Experiments demonstrate correct defocus relationships and natural defocus blur in scenes with glasses and mirrors. Our approach bridges coherent computation and incoherent perception, addressing physical consistency in 3D holography for next-generation human-computer interfaces.

Holographic displays represent a promising approach to next-generation three-dimensional (3D) visualization, capable of reconstructing the wavefronts of 3D scenes^{1–9}. By encoding both amplitude and phase information, holographic displays can reproduce rich depth cues relevant to human visual perception, motivating their use in immersive visualization applications^{10,11}. Compared to stereoscopic or light-field displays, holographic techniques realize 3D scene representation through wavefront reconstruction, which represents a promising path towards immersive visualization.

The key to achieving such visualization lies in computer-generated holography (CGH). CGH represents 3D scenes using geometric primitives such as point clouds, polygonal meshes, or depth layers^{12–17}. Among these, layer-based methods, benefiting from the

accessibility and compact storage of RGB-D data, have been widely adopted for efficient wave propagation and 3D hologram generation. Conventional layer-based algorithms optimize holograms with iterative optimization, which requires extensive computation and time¹⁸. Recent deep learning-based approaches have significantly accelerated hologram generation. Lee et al. synthesize a layer-based dataset to train an end-to-end U-Net that directly predicts holograms from depth-sliced images¹⁹. Wang et al. and Yan et al. propose self-supervised 3D hologram generation networks^{20,21}. However, a common practice among these learning-based methods is to apply supervision only to focused regions during training, neglecting the wavefront fidelity in defocused areas. This leads to abrupt transitions between depth layers and pronounced ringing artifacts in out-of-focus areas.

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The unnatural defocus effect, primarily featured by ringing artifacts, originates from a fundamental conflict in physical models and is exacerbated by technical implementation limitations. On one hand, existing computational methods are based on coherent diffraction models, whereas human visual perception is adapted to the incoherent perception of the natural world. This mismatch between the computational and perceptual models is the primary cause of inaccurate defocus. On the other hand, the combined effect of scene discretization and the finite aperture of optical systems artificially introduces and truncates high-frequency components. This technical limitation further accentuates the ringing artifacts. In the defocused regions, the ringing artifacts appear particularly salient against a smooth background^{22,23}. Since defocus provides critical depth cues for human visual perception²⁴, the inaccurate defocus can severely undermine scene realism and potentially cause visual discomfort^{25–27}.

Several methods have been proposed to improve defocus realism in CGH. Kavakli et al. use Gaussian-blurred images as supervision targets to mimic real-world defocus²³. Yang et al. further improve defocus fidelity by employing a physically based renderer to generate supervision targets for gradient-based optimization²². Akşit et al. and Zhan et al. propose learning-based pipelines, HoloBeam²⁸ and ConfigurableHolography²⁹, which replace iterative optimization with end-to-end neural networks to reduce computation time. Choi et al. and Dong et al. introduce time-multiplexing strategies that enhance perceived 3D realism by inducing inter-frame incoherence^{30,31}, while raising the hardware complexity. Despite their progress, existing methods face several limitations. Iterative and time-multiplexing approaches are prohibitively time-consuming. End-to-end models often struggle to maintain physical consistency, particularly when modeling long-range diffraction, resulting in a loss of fidelity. These limitations underscore the need for further investigation into the physical consistency and realistic defocus effect within a learning-based pipeline.

In this work, we propose a physics-informed, non-iterative methodology for 3D hologram generation with realistic defocus effects, as illustrated in Figs. 1a, b and 2. The central contribution of our work is a representation-consistent holographic framework that aligns complex scene encoding, neural inference, and wave-optical propagation under a unified physical model of defocus formation. Specifically, we begin with a theoretical analysis of human 3D perception, revealing the essential role of the realistic defocus effect for natural depth interpretation. To overcome inaccurate wavefront information in defocused areas, we propose a network that synergistically integrates a neural renderer and a multi-branch phase prediction module to generate physically consistent complex-valued light fields. The light diffraction procedures and the effects of a limited aperture and spatial frequency are modeled to ensure physical consistency with the coherent holographic display system. Furthermore, to address the depth ambiguity of RGB-D data in complex scenes with transparencies or reflections, instead of the conventional method of assigning a single depth value to each pixel, we propose a layered depth image (LDI)-based rendering strategy, as shown in Fig. 1d, that preserves the multi-layer depth structure and ensures correct focus ordering. The predicted wavefront is then physically back-propagated to the spatial light modulator (SLM) plane for phase-only encoding. The design of the multi-branch structures combines RGB-D and amplitude-phase data, leveraging their complementary physical semantics to enhance hologram generation. Our simulations achieve an average peak signal-to-noise ratio (PSNR) of 32.35 dB and an average structural similarity index (SSIM) of 0.938. Our optical experiments demonstrate faithful 3D reconstructions with smooth depth transitions and a photorealistic defocus effect. In addition, the proposed framework can be easily extended to multi-view holographic displays, demonstrating its scalability and generality. The details are in Supplementary Note 7. Together, these results highlight the potential of our physics-informed hologram generation for immersive 3D visualization.

Results

Physics-informed framework for photorealistic 3D holographic displays

Accurate defocus effects are critical for achieving photorealistic depth perception in 3D holographic displays. To address degraded defocus effects and incorrect focus ordering arising from the mismatch between computational and physical models, as well as the limited representational capacity of RGB-D, we propose a physics-informed 3D holographic approach that essentially aligns complex scenes encoding, neural inference, and wave-optical propagation within a unified physical model of defocus formation, constituting the core contribution of this work. Specifically, our LDI-based multi-layer data representation preserves the full visible geometry of scenes containing transparency or reflection. Realistic defocus synthesized through the LDI-based multi-layer rendering provides an incoherent perceptual prior, while explicit wave-optical propagation modeling imposes a physically grounded constraint on the inferred complex-valued fields. Notably, physics-informed in our context denotes the explicit integration of wave-optical propagation into the learning pipeline as a constraint on the neural inference of complex-valued fields, rather than merely using propagation as a post-processing step.

In our setup, a 3D scene is located at a finite distance from the SLM and occupies a volumetric region in space. A conventional single-depth RGB-D representation is physically insufficient to describe incoherent defocus formation, as multiple depth contributions along a single line of sight jointly determine the observed blur. With LDI, we decompose the scene into two semantic layers: a foreground layer capturing the visible appearance of transparent or reflective objects, and a background layer representing either the physical scene geometry behind transparent objects or the virtual scene formed by mirror reflections in reflective surfaces. In the special case where transparency and reflection are absent, this two-layer representation naturally degenerates into a single-layer RGB-D representation.

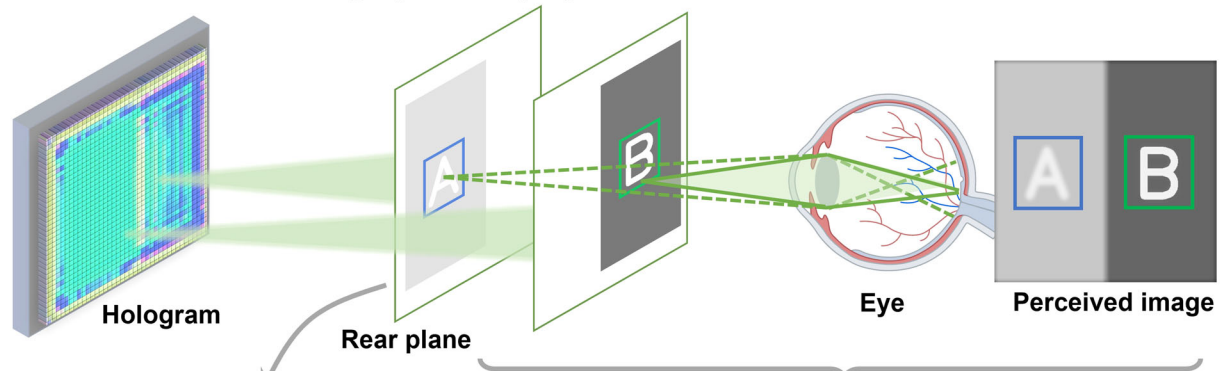
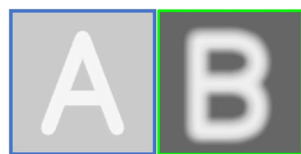
The overview of the proposed physics-informed framework for 3D holographic display is shown in Fig. 2. Each layer consists of an all-in-focus image $x \in \mathbb{R}^{H \times W \times C}$ and a depth map $d \in \mathbb{R}^{H \times W}$. A neural renderer (NR)³² synthesizes a focal stack from the RGB-D input, while a phase generator (PG) predicts the phase of the complex w middle focal plane:

$$\{I_0, \dots, I_{\text{mid}}, \dots, I_{N-1}\} = \text{NR}(x, d; \theta_{\text{NR}}),$$

$$C_{\text{mid}} = \sqrt{I_{\text{mid}}} \cdot e^{j \cdot \text{PG}(x, d; \theta_{\text{PG}})}, \quad (1)$$

where C_{mid} is the complex-valued wavefront at the middle plane³³, with amplitude from the neural renderer $\text{NR}(\cdot, \cdot; \theta_{\text{NR}})$ with parameters of θ_{NR} and phase from the phase generator $\text{PG}(\cdot, \cdot; \theta_{\text{PG}})$ with parameters of θ_{PG} , the $I_{\text{stack}} = \{I_0, \dots, I_{\text{mid}}, \dots, I_{N-1}\}$ is a focal stack of one layer generated by the neural render. The middle plane is defined as $\text{mid} = \lfloor N/2 \rfloor - 1$, where $\lfloor \cdot \rfloor$ rounds down the input to the nearest integer.

Conventional RGB-D data cannot represent cases where a single pixel corresponds to multiple depth values. To address the depth challenge in complex scenes containing transparent or reflective surfaces, we propose a dual-layer RGB-D representation method. Then, we introduce a method for solving the complex-valued light field at the intermediate layer required for multi-depth holographic display. Two focal stacks are rendered from the two RGB-D data layers respectively. The information from these focal stacks is fused via alpha blending. Here, the neural renderer serves two functions: obtaining the complex-valued field of the intermediate layer and supervising the reconstruction of images at multiple depths. With conventional methods, the amplitude of the middle wavefront is inaccessible. Specifically, only the focusing areas can be obtained with the discrete slicing of the 3D scene, while the blur behavior of the defocusing areas

(a) Photorealistic 3D holographic display**(b) Defocus behavior**

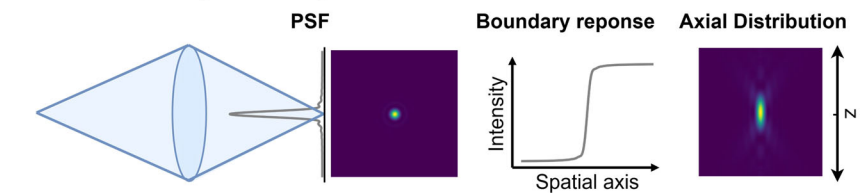
Natural defocus blur



Unrealistic defocus blur

(c) Visual perception analysis

Incoherent case (real-world)



Coherent case (holographic display)

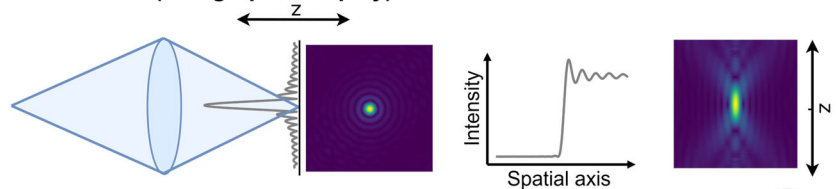
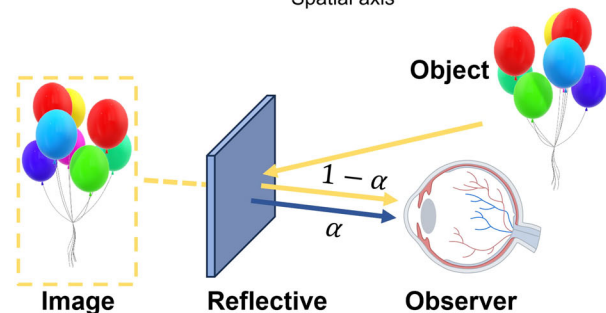
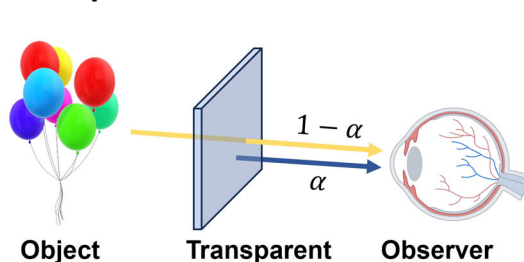
**(d) Transparent and reflective scene**

Fig. 1 | Overview of our 3D holographic display and analysis. **a** The proposed 3D holographic display. **b** The analysis of defocus behavior. **c** Optical properties of coherent and incoherent imaging systems. **d** 3D display of complex scenes with

transparent and reflective surfaces. The parameter α denotes the contribution ratio of the transparent or reflective medium to the final image.

is missing. Neural rendering replenishes the defocus blur in these areas, allowing the middle focal image to serve as the amplitude of the middle-plane wavefront.

The complex-valued field C_{mid} of the middle plane is back-propagated to the SLM plane to obtain the field C_{SLM} . Due to device constraints and diffraction efficiency, we employ a phase-only SLM to load holograms. The complex hologram is encoded into a phase-only hologram H using a phase-only encoder $\text{POE}(\cdot; \theta_{\text{POE}})$ with parameters of θ_{POE} .

$$\begin{aligned} C_{\text{SLM}} &= f(C_{\text{mid}}, -z_{\text{mid}}), \\ H &= \text{POE}(C_{\text{SLM}}; \theta_{\text{POE}}), \\ \hat{I}_{\text{stack}} &= f^2(e^{iH}, z_{\text{stack}}), \end{aligned} \quad (2)$$

where the distances of the focal images are $z_{\text{stack}} = \{z_0, \dots, z_{\text{mid}}, \dots, z_{N-1}\}$, z_{mid} is the distance of the middle plane to the SLM plane, the reconstructed focal stack is $\hat{I}_{\text{stack}} = \{\hat{I}_0, \dots, \hat{I}_{\text{mid}}, \dots, \hat{I}_{N-1}\}$, $f(\cdot, \cdot)$ is the angular spectrum method (ASM)³⁴.

Importantly, in our pipeline, neither the PG nor the POE learn the diffraction process. Instead, diffraction is explicitly modeled and enforced as a physical constraint, while the networks are responsible for inferring the missing components of the complex optical field under this wave-optical formulation. This design ensures that neural inference remains physically grounded rather than implicitly approximating propagation behavior.

The network architectures incorporate domain-specific adaptations for 3D holography. The PG and POE utilize YNet-based architectures³⁵ with separate encoders, capturing distinct semantic features.

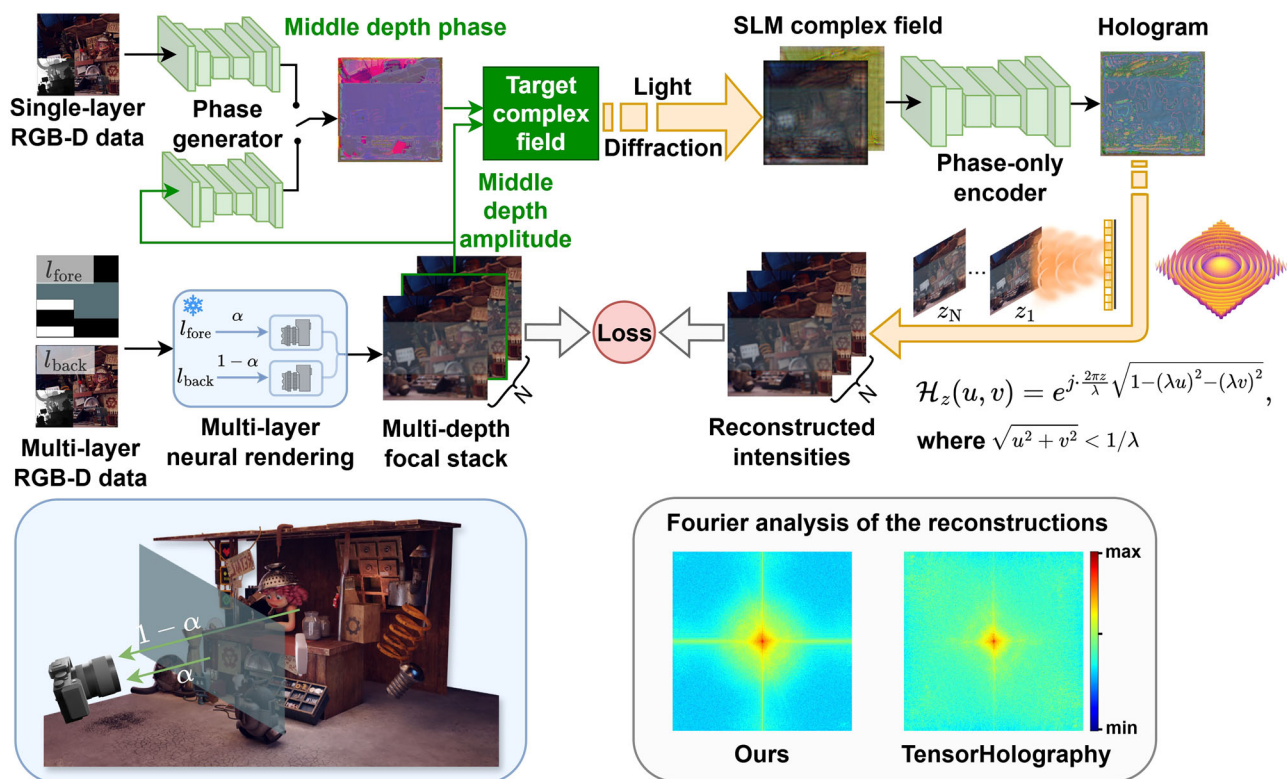


Fig. 2 | Overview of the proposed physics-informed framework for photo-realistic 3D holographic displays. The Fourier analysis of the reconstructions indicates that our result shows a low-frequency-dominated spectrum, which aligns with the spectral feature of the target natural image. In contrast, TensorHolography

artificially introduces high-frequency components. The example data, Junk Shop by Alex Treviño, are licensed under Creative Commons Attribution-ShareAlike (CC BY-SA).

The overall loss function L for training the framework is given by

$$\mathcal{L} = \frac{1}{N} \sum_{n=0}^{N-1} \left(L_2(\hat{I}_n, I_n) + 0.5 * L_2(\nabla \hat{I}_n, \nabla I_n) \right), \quad (3)$$

where $\nabla \cdot$ denotes the first-order gradient operator, $L_2(\cdot, \cdot)$ is the mean square error, and N is the number of focal images.

By inferring physically consistent complex-valued fields under coherent propagation constraints while supervising the network using perception-consistent defocus statistics, the proposed framework reconciles coherent wavefront computation with incoherent defocus perception, realizing a realistic 3D holographic display with correct depth ordering. Additional details of the networks are provided in Supplementary Notes 1, 6, and 9.

Simulation results of realistic defocus effect

Simulation results and average performance are presented in Fig. 3 and Table 1, comparing Multiplane SGD⁴⁸, TensorHolography³³, FS SGD²³, and HoloBeam²⁸. Figure 3 compares the visualized reconstructions of Multiplane SGD, TensorHolography, and our proposed method. The NF and RF denote near focus and rear focus. Multiplane SGD produces clear, speckle-free reconstructions in the focused regions but suffers from severe speckle noise in defocus areas. Additionally, due to the depth slicing of the scene, unnatural ringing artifacts appear at the edges of the sliced planes and in defocused areas. TensorHolography has ringing artifacts in the defocused area³⁶. The Bunny image achieves a PSNR of 24.7 dB and an SSIM of 0.68 in the rear-focus case and a PSNR of 25.1 dB and an SSIM of 0.71 in the near-focus case. Among these results, our method achieves the best performance in both realism and per-pixel metrics. Smooth transitions are achieved across different depth planes. Additionally, the focus areas exhibit high-

quality reconstructions. In the Bunny scene, our method achieves a PSNR of 29.7 dB and an SSIM of 0.92 in the rear-focus case and a PSNR of 30.0 dB and an SSIM of 0.91 in the near-focus case. Additional simulation results comparing Multiplane SGD, TensorHolography, FS SGD, and HoloBeam are provided in Supplementary Note 2.

Table 1 presents the average performance on the test dataset³³. Multiplane SGD has an average PSNR of 23.05 dB and an average SSIM of 0.534. TensorHolography shows 27.91 dB in PSNR and 0.832 in SSIM. FS SGD outperforms Multiplane SGD, achieving PSNR/SSIM of 28.95/0.868, demonstrating that incorporating defocus guidance can improve reconstruction fidelity. Meanwhile, HoloBeam has a PSNR/SSIM of 23.62/0.696 with the middle depth plane at 0.2 m, with additional results at 0.002 m in Supplementary Note 8. Our method outperforms comparison approaches, achieving a PSNR of 32.35 dB and an SSIM of 0.938. More importantly, it also attains high performance across multiple perceptual metrics, including learned perceptual image patch similarity (LPIPS), deep image structure and texture similarity (DISTS), visual information fidelity (VIF), and feature similarity index measure with chromatic extension (FSIM and FSIMc), indicating improved alignment with human visual perception. Our approach leverages realistic defocus priors and a physics-informed pipeline, enabling high-fidelity reconstructions without iterative optimization. Importantly, this performance is obtained with only 64 ms runtime on an Nvidia RTX 3090 GPU. A detailed complexity comparison of different approaches is provided in Supplementary Note 4.

Experimental results of realistic defocus effect

Figure 4b presents the optical reconstructions of Multiplane SGD, TensorHolography, and our proposed method using the benchtop prototype. For Multiplane SGD, the defocused regions of the

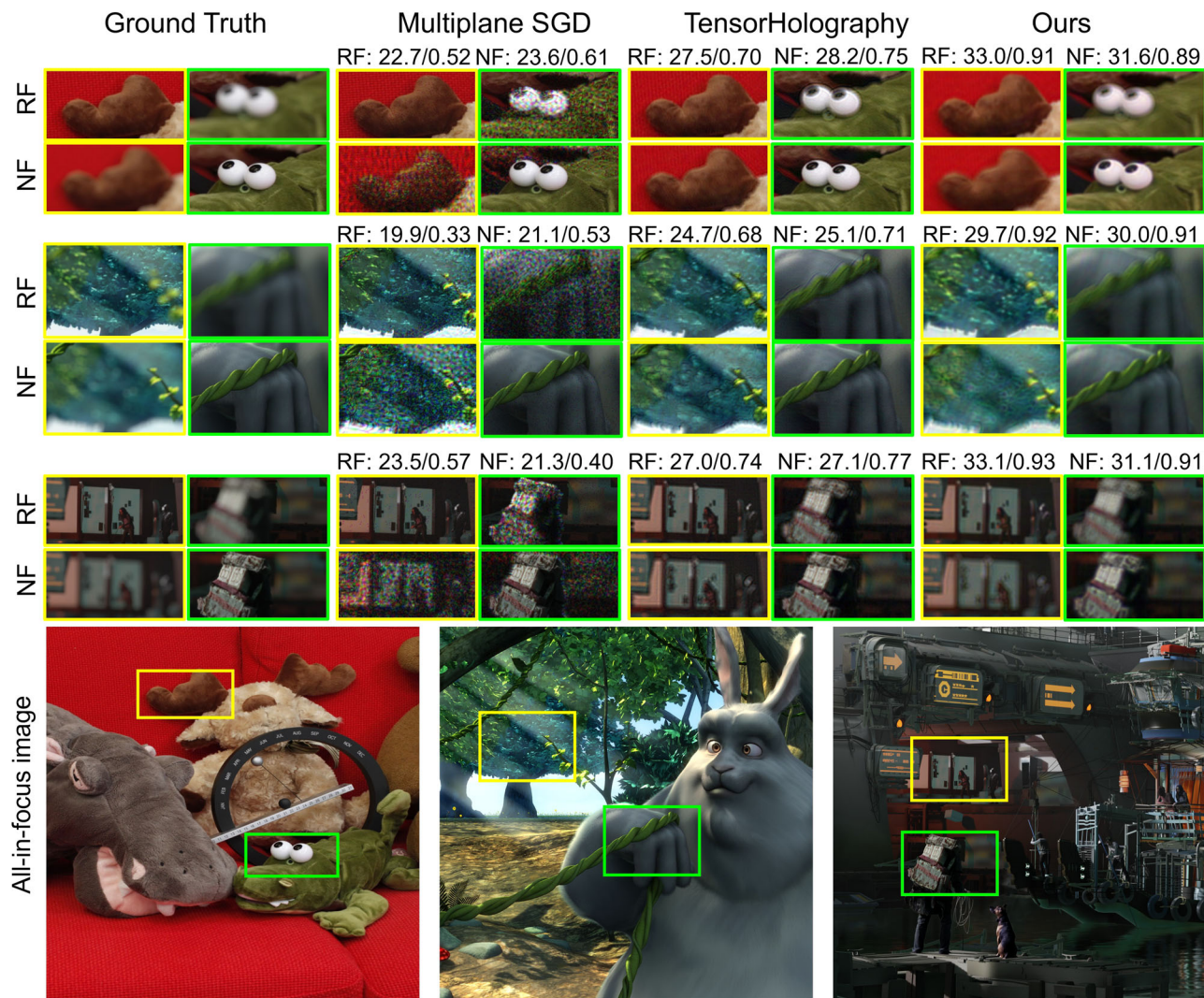


Fig. 3 | Simulation results of Multiplane stochastic gradient descent (SGD)¹⁸, TensorHolography³³, and our method. NF and RF denote near-focus and rear-focus cases. The PSNR (dB) and SSIM values are reported at the top of the enlarged reconstructions. The Couch data are from the Couch dataset⁴⁴. The Bunny data are from Big Buck Bunny, Blender Foundation, Copyright (C) 2008, licensed under Creative Commons Attribution 3.0 (CC-BY 3.0), available at <http://www.bigbuckbunny.org/>. The PartyTug 6:00AM data are from Blender 2.83 Splash Screen by Ian Hubert, licensed under Creative Commons Attribution-ShareAlike 3.0 (CC-BY-SA 3.0).

Table 1 | Quantitative comparison of hologram-generation methods on 100 resized samples from the MIT-CGH-4K dataset³³. The best performance is marked in bold

Methods	Defocus-aware	PSNR/SSIM↑	LPIS↓	DISTS↓	VIF↑	FSIM/FSIMc↑
Multiplane SGD ¹⁸	x	23.05/0.534	0.6405	0.2072	0.4178	0.9135/0.9065
TensorHolography ³³	x	27.91/0.832	0.2833	0.1440	0.5206	0.9127/0.9107
FS SGD ²³	✓	28.95/0.868	0.2873	0.1695	0.5358	0.9443/0.9435
HoloBeam ²⁸	✓	23.62/0.696	0.7100	0.4225	0.2273	0.8360/0.8218
Ours	✓	32.35/0.938	0.1339	0.0523	0.7162	0.9765/0.9753

reconstructions are dominated by speckle noise. For TensorHolography, unnatural ringing artifacts appear along the edges of objects in the defocused regions. Among these methods, our method achieves the most natural and high-quality reconstructions in both the focused and defocused regions. The optical reconstructions demonstrate that our approach preserves the natural depth-dependent blur in defocused areas while maintaining high fidelity reconstructions in the focused regions. Additional experimental results comparing Multiplane SGD, TensorHolography, FS SGD, and HoloBeam are provided in Supplementary Note 3.

Evaluation on transparent and reflective scenes
We construct two representative transparent and reflective scenes, illustrated in Fig. 5a, b. A semi-transparent blue plate with a text Render is placed in front of a Bamboo scene, and a reflective framed mirror is placed in front of a Bunny scene with a virtual reflected Bamboo scene inside the mirror. For the RGB-D representation, occluded or reflected portions are projected onto a 2D transparent or reflective surface, resulting in the loss of depth information. In contrast, our LDI-based multi-layer representation preserves the depth map of the occluded objects in the background layer.

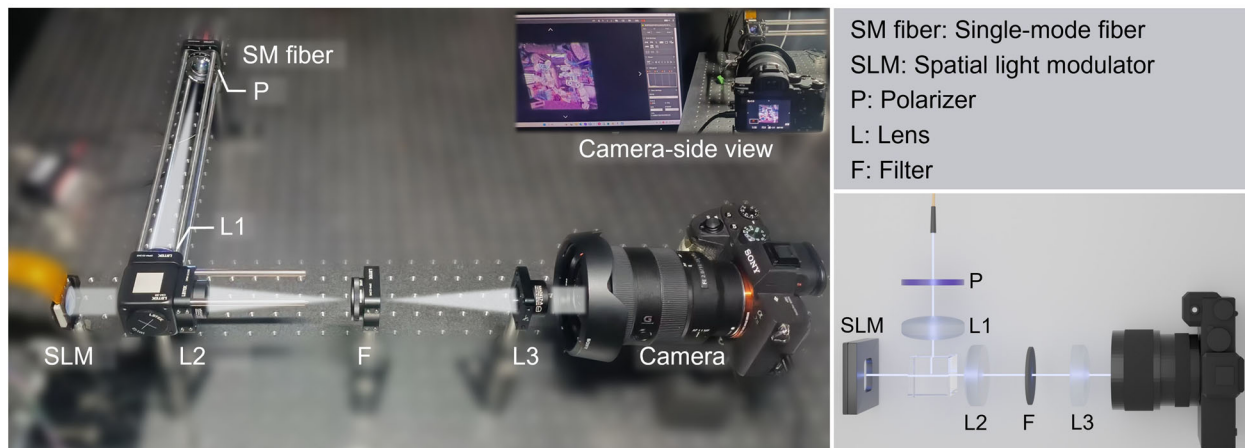
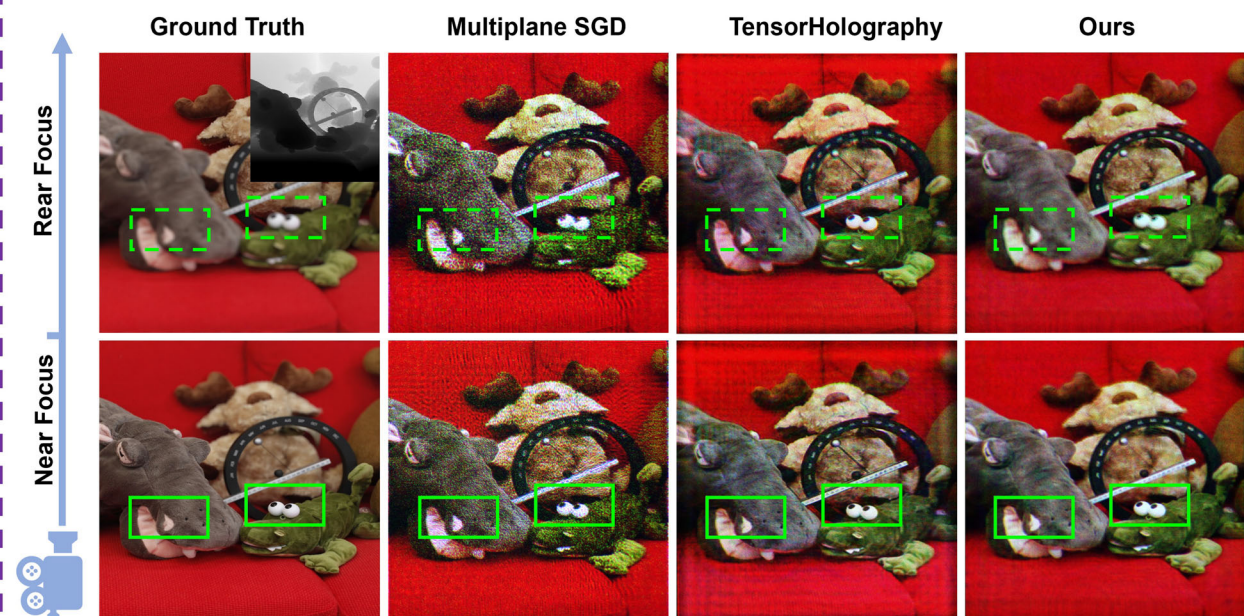
(a) Experimental setup**(b) Experimental results**

Fig. 4 | Experimental setup and results. a Optical setup of the holographic display experiment. **b** Experimental results of Multiplane SGD¹⁸, TensorHolography³³, and the proposed method. The solid boxes represent the focused areas, and the dashed boxes represent the defocused areas. The Couch data are from the Couch dataset⁴⁴.

The experimental results in Fig. 5c validate our analysis. In the rear-focus setting of the transparent scene, our LDI-based multi-layer pipeline preserves the correct depth ordering. The background leaves and patches of sky visible through the bamboo appear sharp while the foreground elements are blurred. In contrast, the RGB-D baseline incorrectly blurs the background leaves and sky. A similar phenomenon is observed in the reflective scene. RGB-D baseline erroneously blurs the background human in the rear-focus setting, whereas our approach maintains correct focus behavior. Detailed analysis and additional results are presented in Supplementary Note 6 of the Supplementary Information.

Discussion

Accurate focus and defocus cues are critical for natural 3D perception. However, existing 3D holographic displays based on RGB-D representations and coherent illumination models often exhibit unnatural

ringing artifacts and incorrect depth ordering, especially for scenes containing transparency or reflection.

To overcome these challenges, we propose a physics-informed holographic framework that generates physically consistent wavefronts for complex 3D scenes. The core contribution of our framework lies in the alignment of complex 3D scenes encoding, neural inference, and wave-optical propagation under a unified physical model of defocus formation. Specifically, our approach employs an LDI-based multi-layer rendering strategy to preserve multi-layer scene structure and resolve depth ambiguity beyond the constraints of single-depth RGB-D representations. By explicitly embedding a diffraction model into the learning pipeline, the framework models coherent propagation, limited aperture, and spatial-frequency effects. Furthermore, the pipeline naturally extends to multi-view holographic displays, demonstrating both scalability and generality.

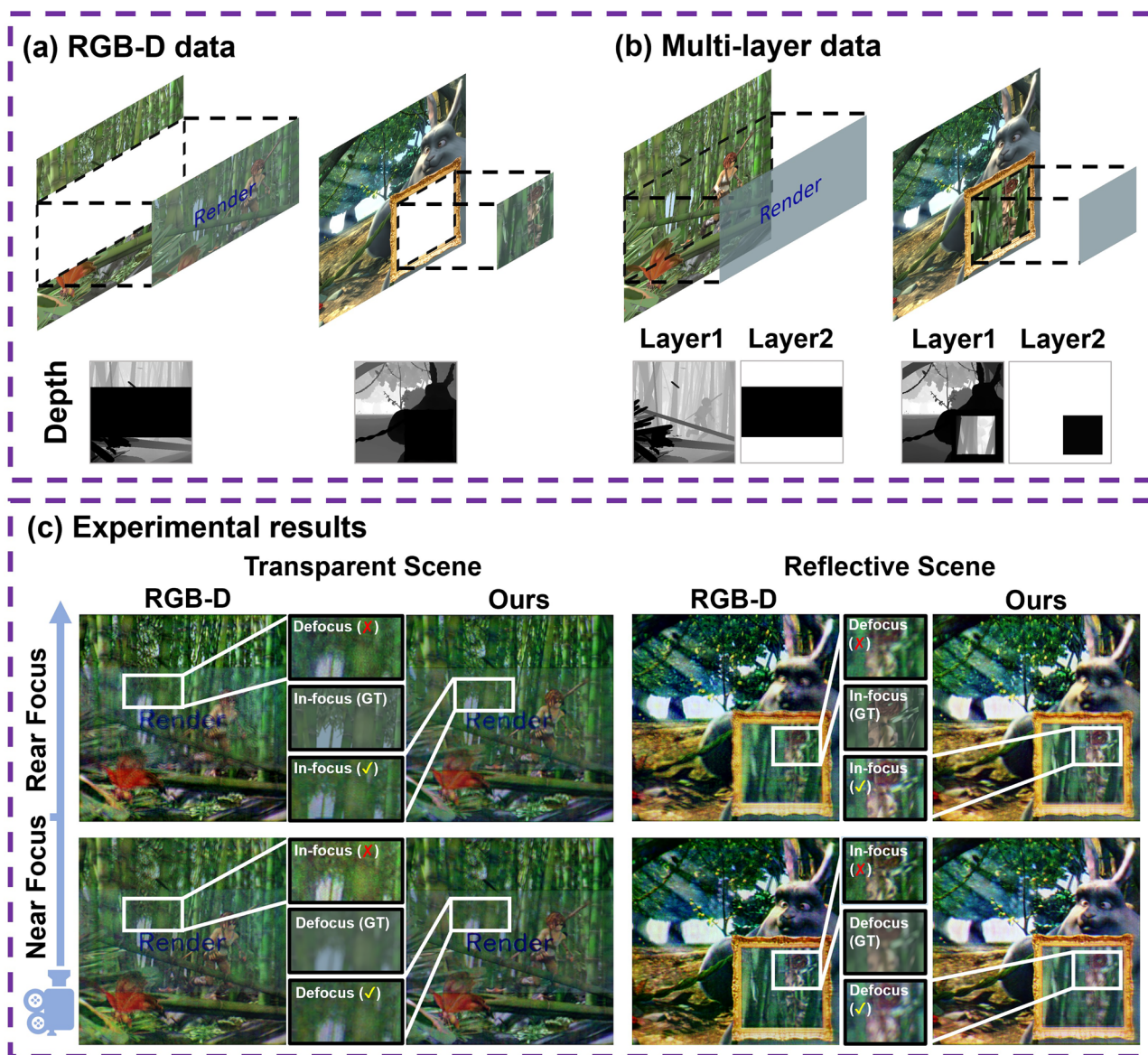


Fig. 5 | Experimental results on scenes with transparency or reflection. **a** Illustration of RGB-D data representation. **b** Proposed LDI-based multi-layer representation for 3D scenes containing transparency and reflection. **c** Experimental results on transparent and reflective scenes, comparing the proposed LDI-based multi-layer pipeline with the RGB-D baseline. Reconstructed areas

at incorrect and correct depths are marked with a cross and check marks, respectively. The Bamboo data are from the MPI-Sintel dataset⁴⁵. The Bunny data are from Big Buck Bunny, Blender Foundation, Copyright (C) 2008, licensed under Creative Commons Attribution 3.0 (CC-BY 3.0), available at <http://www.bigbuckbunny.org/>.

Extensive simulations and optical experiments validate the effectiveness of the proposed method, showing photorealistic defocus effects and correct depth ordering. Our work addresses a long-standing inconsistency between coherent wavefront computation and incoherent visual perception in 3D holographic displays, and establishes a principled pathway toward physically consistent and perceptually faithful 3D visualization.

Methods

The Analysis of Defocus Blur Perception Under Coherent and Incoherent Illumination

Holographic displays reconstruct wavefronts via diffraction, typically using coherent light sources. The stable phase of coherent light enables the recording of high-contrast interference patterns necessary for holography^{37,38}. However, the human visual system has evolved to process 3D scenes illuminated by the incoherent natural light, which is

composed of light with random phase and a broad frequency spectrum.

A major depth cue of human 3D vision is the defocus blur. Under incoherent illumination of natural light, defocus blur typically exhibits a smooth profile³⁹. Under coherent illumination, computer-generated holography usually suffers from ringing artifacts, especially in defocused regions. To mathematically investigate the causes of this artifact, we analyze a simplified eye-imaging system comprising a thin convex lens and a finite aperture, as shown in Fig. 1c. The point spread function (PSF) of a coherent imaging system $h_c(x, y)$ is the Fourier transform of the pupil function $P(\cdot, \cdot)$ ⁴⁰.

$$h_c(x, y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} P(\lambda d_0 u, \lambda d_0 v) e^{-j2\pi(xu + yv)} du dv, \quad (4)$$

where $r = \sqrt{u^2 + v^2}$ as the polar coordinate, λ is the wavelength, u and v are the spatial frequencies, and d_0 is the distance from the lens to the imaging plane.

Suppose the aperture is circular, the coherent PSF of the imaging system is proportional to the Fourier transform of a circular function $\text{circ}(r)$ ³⁴.

$$\mathcal{F}\{\text{circ}(r)\} = 2\pi \int_0^R r J_0(2\pi r \rho) dr = 2\pi R^2 \frac{J_1(2\pi R \rho)}{2\pi R \rho}, \quad (5)$$

where r and ρ are the variables of polar coordinates, R is the radius of the circular function, and $\mathcal{F}\{\cdot\}$ denotes the 2D Fourier transform. J_0 and J_1 are the order zero and order one Bessel functions. The PSF of an incoherent system is the product of $h_c(x, y)$ and its complex conjugation:

$$h_i(x, y) = h_c(x, y) \cdot h_c^*(x, y). \quad (6)$$

The line spread function (LSF) $L(x)$ and the edge spread function (ESF) $E(x)$ influence the perception of edges in images. They are defined as the responses to line impulse and step functions.

$$E(x) = \int_{-\infty}^x L(\alpha) d\alpha = \int_{-\infty}^x \int_{-\infty}^{+\infty} h(\alpha, \beta) d\beta d\alpha, \quad (7)$$

where h represents h_c or h_i . The defocus PSF $h_{\text{def}}(x, y)$ is given by

$$h_{\text{def}}(x, y) = f(h_c(x, y), z_{\text{def}}), \quad (8)$$

where $f(\cdot, \cdot)$ is the ASM³⁴, and z_{def} is the defocus distance. The ASM for propagating a complex-valued field of $U(x, y)$ by a distance of z is given by

$$f(U(x, y), z) = \iint_{-\infty}^{+\infty} \mathcal{F}\{U(x, y)\} \mathcal{H}(u, v) e^{j2\pi(ux + vy)} dudv, \quad (9)$$

$$\mathcal{H}(u, v) = \begin{cases} e^{j\frac{\pi}{\lambda} \sqrt{1 - (\lambda u)^2 - (\lambda v)^2}}, & \sqrt{u^2 + v^2} < \frac{1}{\lambda} \\ 0, & \text{otherwise} \end{cases}$$

The incoherent defocus PSF is the product of the coherent defocus PSF and its complex conjugate. Based on the above equations, we can analyze the effects of coherent and incoherent illumination on human 3D depth perception. As shown in Fig. 1b, c, the coherent PSF oscillates near the edges of objects, introducing undesirable ringing artifacts that are uncommon in natural 3D vision. Additionally, the axial distribution of the PSF along the optical axis reveals that, for coherent illumination, the defocus PSF shows pronounced ringing effects as the object deviates from the focal plane. In contrast, the incoherent defocus PSF exhibits a much smoother profile, with gradual transitions that better align with human visual expectations⁴¹. Algorithms for generating coherent holograms with natural incoherent defocus effects remain to be investigated.

Simulation settings

To assess the proposed approach, we conduct comprehensive simulations with Multiplane SGD¹⁸, TensorHolography³³, FS SGD²³, and HoloBeam²⁸ for comparison. Multiplane SGD and TensorHolography are categorized as defocus-unaware methods. Multiplane SGD supervises the focused regions at multiple target depths and optimizes holograms. TensorHolography predicts middle-complex-valued holograms using a network and subsequently encodes them on the SLM plane as phase-only patterns via the anti-aliasing double-phase method. In contrast, FS SGD and HoloBeam are categorized as defocus-aware methods. FS SGD supervises reconstructions across multiple planes, where defocused regions are synthesized using Gaussian-kernel blurs, thereby providing defocus guidance. HoloBeam

employs an end-to-end UNet to generate phase-only holograms, trained on data produced by FS SGD.

The training dataset is from DeepFocus³². The approaches are tested on the resized MIT-CGH-4K dataset³³. The focal images are distributed evenly across 8 planes, ranging from 19.7 cm to 20.3 cm. The depth range across the 3D scene is 6 mm, which is consistent with TensorHolography. Our models are implemented in Python 3.10.14 and PyTorch 2.5.1⁴². All algorithms are trained and tested on a NVIDIA GeForce RTX 3090 GPU. The proposed network is trained for 200 epochs with a learning rate of 3×10^{-4} . Adam is used to optimize the networks⁴³. SGD runs for 500 iterations to ensure adequate convergence. The pixel pitch is $6.4 \mu\text{m}$, and the wavelengths are 638 nm, 520 nm, and 450 nm. The numerical reconstructions are obtained via ASM.

Experimental details

We built a benchtop prototype to validate the proposed framework, as illustrated in Fig. 4a. The incident light is linearly polarized, collimated, modulated by the SLM, and filtered using a 4f system. The FISBA READYBeam laser module operates at wavelengths of 450 nm, 520 nm, and 638 nm. A polarizer ensures a linearly polarized incident beam, which is collimated by a lens. A beam splitter divides the beam into two paths. The incident beam is modulated by a HOLOEYE LETO-3-CFS-127 SLM with a pixel pitch of $6.4 \mu\text{m}$. The modulated beam is reflected, and an aperture filters out unwanted high-order diffraction. Monochromatic reconstructions are captured separately for each wavelength and combined to produce the final color reconstructions.

Data availability

The data supporting the findings of this study are provided in the main text, the Supplementary Information, and GitHub, <https://github.com/THUIntelligentOpticsLab/NaturalDefocus>.

Code availability

The code supporting the findings of this study is available at the GitHub, <https://github.com/THUIntelligentOpticsLab/NaturalDefocus>.

References

- Chang, C., Bang, K., Wetzstein, G., Lee, B. & Gao, L. Toward the next-generation VR/AR optics: A review of holographic near-eye displays from a human-centric perspective. *Optica* **7**, 1563–1578 (2020).
- Yuan, G., Zhou, M., Peng, Y., Chen, M. & Geng, Z. Error-compensation network for ringing artifact reduction in holographic displays. *Opt. Lett.* **49**, 3210–3213 (2024).
- Liu, Z. et al. Comprehensive compensation of real-world degradations for robust single-pixel imaging. *Light.: Sci. Appl.* **14**, 365 (2025).
- Yuan, G. et al. Physics-aware cross-domain fusion aids learning-driven computer-generated holography. *Photonics Res.* **12**, 2747–2756 (2024).
- Wei, Y., Zhu, G., Yuan, G., Zhou, M. & Geng, Z. Text-to-hologram: Hologram generation from natural language. *Proc. SPIE Fifth Int. Computational Imaging Conf. (CITA 2025)* **14000**, 652–658 (2026).
- Luo, X. et al. Revolutionizing optical imaging: Computational imaging via deep learning. *Photonics Insights* **4**, R03–R03 (2025).
- M. Zhou, H. Zhang, M. K. Chen, and Z. Geng, Implicit feature compression for efficient cloud-edge holographic display, *Displays*, 103151, (2025).
- Wei, Y. et al. Speckle-free holography with a diffraction-aware global perceptual model. *Photonics Res.* **12**, 2418–2423 (2024).
- Zhou, M., Zhang, H., Jiao, S., Chakravarthula, P. & Geng, Z. End-to-end compression-aware computer-generated holography. *Opt. Express* **31**, 43908–43919 (2023).

10. Sahin, E., Stoykova, E., Mäkinen, J. & Gotchev, A. Computer-generated holograms for 3D imaging: A survey. *ACM Comput. Surv. (CSUR)* **53**, 1–35 (2020).
11. Song, Y. et al. Three-dimensional varifocal meta-device for augmented reality display. *Photonix* **6**, 6 (2025).
12. Tsang, P., Poon, T. C. & Wu, Y. Review of fast methods for point-based computer-generated holography. *Photonics Res.* **6**, 837–846 (2018).
13. Chen, R. H. Y. & Wilkinson, T. D. Computer generated hologram from point cloud using graphics processor. *Appl. Opt.* **48**, 6841–6850 (2009).
14. Matsushima, K. & Nakahara, S. Extremely high-definition full-parallax computer-generated hologram created by the polygon-based method. *Appl. Opt.* **48**, H54–H63 (2009).
15. Zhang, Y. et al. Polygon-based computer-generated holography: A review of fundamentals and recent progress. *Appl. Opt.* **61**, B363–B374 (2022).
16. Chen, J. S. & Chu, D. Improved layer-based method for rapid hologram generation and real-time interactive holographic display applications. *Opt. Express* **23**, 18143–18155 (2015).
17. Zhao, Y., Cao, L., Zhang, H., Kong, D. & Jin, G. Accurate calculation of computer-generated holograms using angular-spectrum layer-oriented method. *Opt. Express* **23**, 25440–25449 (2015).
18. Choi, S., Gopakumar, M., Peng, Y., Kim, J. & Wetzstein, G. Neural 3D holography: Learning accurate wave propagation models for 3D holographic virtual and augmented reality displays. *ACM Trans. Graph. (TOG)* **40**, 1–12 (2021).
19. Lee, J. et al. Deep neural network for multi-depth hologram generation and its training strategy. *Opt. Express* **28**, 27137–27154 (2020).
20. Wang, D. et al. Liquid lens based holographic camera for real 3D scene hologram acquisition using end-to-end physical model-driven network. *Light.: Sci. Appl.* **13**, 62 (2024).
21. Yan, X. et al. Generating multi-depth 3D holograms using a fully convolutional neural network. *Adv. Sci.* **11**, 2308886 (2024).
22. Yang, D. et al. Diffraction-engineered holography: Beyond the depth representation limit of holographic displays. *Nat. Commun.* **13**, 6012 (2022).
23. K. Kavaklı, Y. Itoh, H. Urey, and K. Akşit, Realistic defocus blur for multiplane computer-generated holography, in *Proc. IEEE VR*, 2023, pp. 418–426.
24. Maiello, G., Chessa, M., Solari, F. & Bex, P. J. The (in)effectiveness of simulated blur for depth perception in naturalistic images. *PLOS ONE* **10**, e0140230 (2015).
25. Emoto, M., Niida, T. & Okano, F. Repeated vergence adaptation causes the decline of visual functions in watching stereoscopic television. *J. Disp. Technol.* **1**, 328 (2005).
26. Hoffman, D. M., Girshick, A. R., Akeley, K. & Banks, M. S. Vergence-accommodation conflicts hinder visual performance and cause visual fatigue. *J. Vis.* **8**, 33–33 (2008).
27. Ukai, K. & Howarth, P. A. Visual fatigue caused by viewing stereoscopic motion images: Background, theories, and observations. *Displays* **29**, 106–116 (2008).
28. Akşit, K. & Itoh, Y. Holobeam: Paper-thin near-eye displays, in *Proc. IEEE Conf. Virtual Reality and 3D User Interfaces (VR)*, 581–591 (2023).
29. Zhan, Y., Shi, Matusik, L., Sun, Q. & Akşit, K. Configurable holography: Towards display and scene adaptation, arXiv preprint arXiv:2405.01558, (2024).
30. Choi, S. et al. Time-multiplexed neural holography: A flexible framework for holographic near-eye displays with fast heavily-quantized spatial light modulators, in *ACM SIGGRAPH 2022 Conference Proceedings*, 1–9 (2022).
31. Dong, Z., Ling, Y., Li, Y. & Su, Y. Motion hologram: Jointly optimized hologram generation and motion planning for photorealistic 3D displays via reinforcement learning. *Sci. Adv.* **11**, eads9876 (2025).
32. Xiao, L., Kaplanyan, A., Fix, A., Chapman, M. & Lanman, D. Deep-focus: Learned image synthesis for computational displays. *ACM Trans. Graph. (TOG)* **37**, 1–13 (2018).
33. Shi, L., Li, B., Kim, C., Kellnhöfer, P. & Matusik, W. Towards real-time photorealistic 3D holography with deep neural networks. *Nature* **591**, 234–239 (2021).
34. Goodman, J. W. Introduction to fourier optics. roberts and company publishers, 2005.
35. Farshad, A., Yeganeh, Y., Gehlbach, P., & Navab, N. Y-net: A spatio-spectral dual-encoder network for medical image segmentation, in *MICCAI*, 582–592 (2022).
36. Yan, X. et al. Recoding double-phase holograms with the full convolutional neural network. *Opt. Laser Technol.* **174**, 110667 (2024).
37. Deng, Y. & Chu, D. Coherence properties of different light sources and their effect on the image sharpness and speckle of holographic displays. *Sci. Rep.* **7**, 5893 (2017).
38. Lee, S. et al. Light source optimization for partially coherent holographic displays with consideration of speckle contrast, resolution, and depth of field. *Sci. Rep.* **10**, 18832 (2020).
39. Ward, G. J., Rubinstein, F. M. & Clear, R. D. A ray tracing solution for diffuse interreflection, in *Proc. SIGGRAPH*, 85–92 (1988).
40. Ersoy, O. K. Diffraction, fourier optics and imaging. John Wiley & Sons, (2006).
41. Watt, S. J., Akeley, K., Ernst, M. O. & Banks, M. S. Focus cues affect perceived depth. *J. Vis.* **5**, 7–7 (2005).
42. Paszke, A. et al. PyTorch: An imperative style, high-performance deep learning library, *NeurIPS*, **32**, (2019).
43. Kingma, D. P. & Ba, J. Adam: A method for stochastic optimization, in *Proc. Int. Conf. Learn. Representations (ICLR)* (2015).
44. Kim, C., Zimmer, H., Pritch, Y., Sorkine-Hornung, A. & Gross, M. H. Scene reconstruction from high spatio-angular resolution light fields. *ACM Trans. Graph.* **32**, 73 (2013). 1–12pp.
45. D. J. Butler, J. Wulff, G. B. Stanley, and M. J. Black, A naturalistic open source movie for optical flow evaluation, in *European Conference on Computer Vision (ECCV)*, 611–625 (2012).

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Author contributions

M.Z. and Z.G. contributed equally to this work. M.Z. and Z.G. conceived the research, developed the model, and conducted the simulations. C.M., X.W., J.S., H.W., K.D., F.L., M.K.C., C.Z., and Z.G. contributed to experimental implementation, data analysis, and theoretical comparisons. M.S., L.L., X.W., M.K.C., C.Z., and Z.G. supervised the research, provided technical support, data analysis, and contributed to manuscript preparation. Z.G. guided the conceptual development and provided technical direction throughout the project. All the authors have accepted responsibility for the entire content of this submitted manuscript and approved the submission. All authors analyzed the results and contributed to preparing the manuscript and discussions.

Competing interests

The authors declare no competing interests.

Additional information

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